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MSc Language Analysis and Processing

Evaluating Large Language Models for Clinical Text Restructuring

A Comparative Analysis with Systematic Error Taxonomy

Internship Research Report

Clinical Natural Language Processing

Task:	Spanish Clinical Exam → English Clinical Narrative
Models:	MedLlama-2-7B, Mistral-7B-Instruct, Qwen2.5-1.5B
Dataset:	CC-MIR Spanish Medical Examination Corpus
Evaluations:	216 total (72 per model across 6 prompt architectures)

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Abstract

Background. Large language models show promise for clinical text processing, yet their reliability for transforming structured medical content while preserving factual accuracy remains poorly understood. The task of converting examination-style clinical justifications into coherent narratives requires models to distinguish between *structural transformation* (changing format) and *semantic preservation* (maintaining content specificity).

Methods. We evaluated three LLMs: MedLlama-2-7B (medical specialist), Mistral-7B-Instruct (generalist), and Qwen2.5-1.5B (small-scale baseline). Testing used six prompt architectures with the CC-MIR Spanish medical examination corpus (216 total evaluations). A three-dimensional framework assessed Fidelity, Restructuring Quality, and Clinical Clarity. We developed a systematic error taxonomy from 127 documented failures and conducted novel specificity preservation analysis.

Results. Mistral-7B achieved superior performance (3.61/6, 60%) versus MedLlama (2.56/6, 43%) and Qwen (1.97/6, 33%). Our central finding is that models systematically confuse *structural transformation* with *semantic generalisation*: when instructed to change text format, they simultaneously replace specific clinical terms with vague superordinates. Specificity preservation rates were alarmingly low: Qwen 13%, MedLlama 20%, Mistral 57%. We identified four hallucination categories: lexical fabrication (31%), value distortion (28%), temporal manipulation (24%), and anatomical confusion (17%).

Conclusions. Medical domain pretraining does not improve constrained text transformation; instruction-following capability proves decisive. The structural-semantic confusion phenomenon has implications for all text transformation tasks requiring specificity preservation.

Keywords: Clinical NLP · Hallucination Detection · Structural-Semantic Confusion · Medical Text Processing · Specificity Preservation · LLM Evaluation

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1 Introduction

1.1 Problem Statement

Clinical examination justifications present a distinctive text transformation challenge. Medical licensing examinations such as the Spanish MIR (*Médico Interno Residente*) employ option-based eliminative reasoning: candidates must identify the correct diagnosis by systematically ruling out alternatives based on clinical evidence. These justifications contain rich medical knowledge but in a format unsuitable for clinical documentation or downstream NLP applications.

The transformation task appears straightforward: convert option-based reasoning into unified clinical narratives. However, this task conceals a subtle but critical distinction that models systematically fail to grasp:

The Core Task Distinction

STRUCTURAL Transformation (Required):
- Remove option identifiers ("Option A", "Option B")
- Eliminate eliminative reasoning patterns ("incorrect because...")
- Convert fragmented justifications into flowing prose

SEMANTIC Preservation (Required):
- Maintain exact medical terminology (e.g., `matidez a la percusión`)
- Preserve all quantitative values (e.g., "proteinuria 1.2 g/day")
- Retain temporal qualifiers (e.g., "acute", "within hours")
- Keep anatomical specificity (e.g., "left upper quadrant")

The Problem: Models conflate these independent operations, performing semantic generalisation when only structural transformation is requested.

1.2 Research Questions

This study addresses four research questions:

RQ1 Does medical domain pretraining improve performance on constrained clinical text transformation tasks?

RQ2 What systematic failure patterns emerge when LLMs restructure clinical content?

RQ3 Do models confuse structural transformation with semantic generalisation?

RQ4 How does prompt architecture influence specificity preservation?

1.3 Contributions

This work makes four contributions to clinical NLP:

1. **Discovery of Structural-Semantic Confusion**: We document and quantify a fundamental failure mode where models incorrectly bundle format transformation with content generalisation
2. **Specificity Preservation Metrics**: Novel quantification of how models lose medical specificity through value dropping, term hypernyming, and qualifier loss
3. **Four-Category Error Taxonomy**: Systematic classification of 127 errors into lexical fabrication, value distortion, temporal manipulation, and anatomical confusion
4. **Evidence that Instruction-Following Outweighs Domain Specialisation**: The generalist Mistral outperformed the medical-specialist MedLlama by 41%

2 Related Work

2.1 Clinical Language Models

The development of domain-specific language models for biomedicine has accelerated rapidly. BioBERT (4) demonstrated that continued pretraining on PubMed abstracts improves biomedical NLP tasks. ClinicalBERT (1) extended this to clinical notes from MIMIC-III. PubMedBERT showed that domain-specific vocabulary and pretraining from scratch can outperform mixed-domain approaches.

More recently, large language models have been evaluated for clinical applications. Med-PaLM (6) achieved expert-level performance on medical question answering, whilst GPT-4 demonstrated strong clinical reasoning capabilities. However, these evaluations typically focus on question answering or summarisation rather than constrained text transformation with strict fidelity requirements.

2.2 Hallucination in Medical NLP

Hallucination (the generation of content not grounded in input) poses particular risks in medical contexts. Ji et al. (3) provide a comprehensive taxonomy distinguishing intrinsic hallucinations (contradicting source) from extrinsic hallucinations (unverifiable additions). Castilho et al. (2) documented hallucination patterns in LLM-generated clinical translations, identifying fabricated terminology and temporal distortions.

Our work extends this literature by identifying a previously undocumented hallucination mechanism: *over-generalisation through hypernymic substitution*, where models replace specific clinical terms with vague superordinates during text transformation.

2.3 Text Transformation and Paraphrase Generation

Text transformation tasks require models to change surface form whilst preserving meaning. Prior work has examined style transfer, simplification, and paraphrase generation. However, these tasks typically allow (or even encourage) semantic smoothing and generalisation. Clinical text transformation differs fundamentally: it requires format change with zero semantic modification.

3 Methodology

3.1 Task Definition

Clinical Text Restructuring Task

Input: Spanish medical examination justification with option-based eliminative reasoning
Output: Unified English clinical narrative preserving all medical facts
Hard Constraint: Zero information addition, zero information loss, zero generalisation

The task requires cross-lingual transfer (Spanish → English), structural transformation (exam format → narrative), and strict semantic preservation. This combination makes it particularly challenging and diagnostic of model capabilities.

3.2 Dataset: CC-MIR Corpus

We used the CC-MIR (CasiMedicos-MIR) corpus, derived from Spanish medical licensing examinations. Each case contains:

- Clinical vignette describing patient presentation
- Five diagnostic options with evidence-based justifications
- Argument structure marking which evidence supports or attacks each option
- Correct answer with detailed clinical reasoning

Twelve clinical specialisations were represented: Cardiology, Nephrology, Neurology, Pneumology, Emergency Medicine, Endocrinology, Psychiatry, Gynaecology, Paediatrics, Infectious Disease, Anatomy, and Palliative Care.

3.3 Models Evaluated

Table 1: Model specifications and selection rationale

Model	Parameters	Focus	Selection Rationale
MedLlama-2-7B	7B	Biomedical	Medical specialist; tests whether domain pretraining helps
Mistral-7B-Instruct	7B	Instructions	Strong instruction-follower; tests task compliance over domain knowledge
Qwen2.5-1.5B	1.5B	General	Small-scale baseline; stress-tests minimum viable capability

3.4 Prompt Architectures

Six prompt designs were evaluated, representing different instruction-following paradigms:

Table 2: Prompt architecture descriptions and performance summary

ID	Architecture	Description	Mean Score
P1	One-Shot Demo	System prompt with worked example	1.0/6
P2	Zero-Shot Instruction	Pure instruction with explicit constraints	**4.5/6**
P3	Delimiter-Based	Structural markers (###) for sections	3.8/6
P4	Do/Don't Constraints	Explicit permitted/forbidden behaviours	3.2/6
P5	Anti-CoT	Direct output without reasoning steps	1.5/6
P6	Output-Targeted	Specified length (5 to 8 sentences)	2.8/6

3.5 Evaluation Framework

3.5.1 Three-Dimensional Scoring

Each output received scores on three dimensions (0 to 2 each, 6 total):

Table 3: Operationalised scoring rubric with explicit criteria

Dim.	Score	Criteria
Fidelity	0	Two or more fabricated facts, OR key medical fact omitted, OR invented terminology
	1	One minor error (modifier change, non-critical value alteration)
	2	Complete preservation of all medical facts; no hallucinations
Restruct.	0	Exam structure fully retained (option numbers, "incorrect because...")
	1	Partial transformation; some eliminative reasoning remains
	2	Complete conversion to unified clinical narrative
Clarity	0	Logical contradictions or incoherent structure
	1	Coherent but awkward clinical flow
	2	Natural clinical discourse appropriate for documentation

3.5.2 Specificity Preservation Analysis

Beyond the three-dimensional framework, we developed metrics to capture the **structural-semantic confusion** phenomenon:

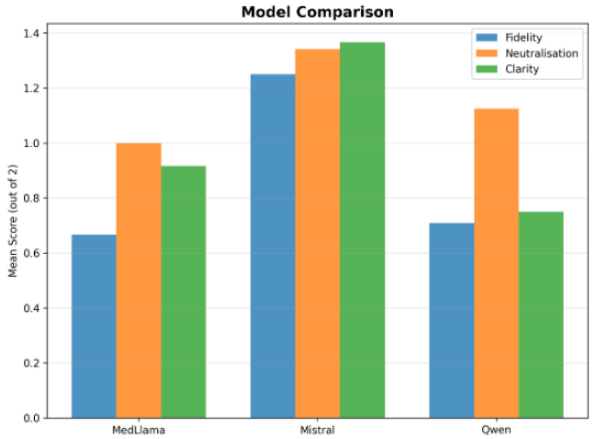
- **Values Dropped**: Quantitative information omitted entirely
- **Values Generalised**: Specific values replaced with qualitative descriptors
- **Terms Hypernymed**: Specific diagnoses/findings replaced with superordinates
- **Qualifiers Lost**: Temporal, severity, or anatomical modifiers dropped
- **Preserved Correctly**: Specific terms maintained exactly as in source

4 Results

4.1 Overall Model Performance

Table 4: Performance summary across all evaluations (n=72 per model)

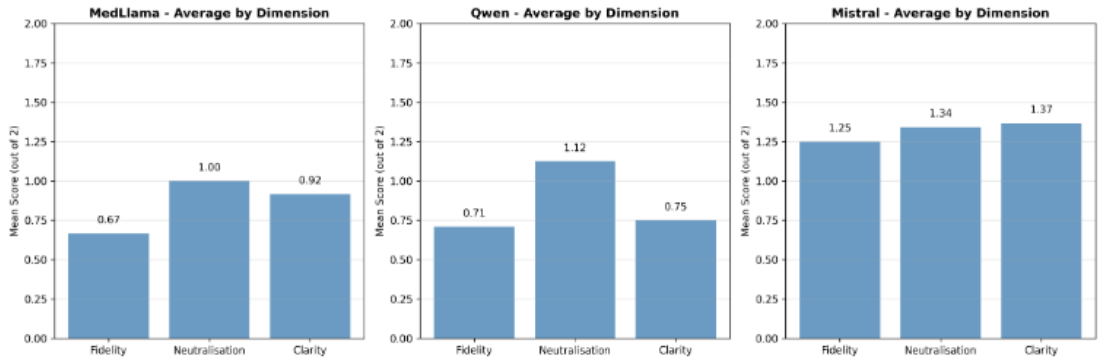
Model	Composite	Fidelity	Restruct.	Clarity	Halluc.
Mistral-7B	**3.61 ± 1.14**	1.06 ± 0.64	1.50 ± 0.50	1.06 ± 0.23	**37%**
MedLlama-2-7B	2.56 ± 1.17	0.53 ± 0.60	1.06 ± 0.64	0.97 ± 0.16	67%
Qwen2.5-1.5B	1.97 ± 1.91	0.44 ± 0.81	1.00 ± 0.67	0.53 ± 0.62	64%



Bar chart showing Mistral's superior performance across Fidelity (1.25), Neutralisation (1.34), and Clarity (1.37) dimensions

Figure 1: Model comparison across the three evaluation dimensions. Mistral leads across all dimensions (Fidelity: 1.25, Neutralisation: 1.34, Clarity: 1.37) compared to MedLlama and Qwen.

Figure 3: Dimensional Score Comparison Across Models



Three-panel comparison showing Mistral leads across all dimensions (Fidelity: 1.25, Neutralisation: 1.34, Clarity: 1.37) compared to MedLlama and Qwen

Figure 2: Three-panel comparison showing dimensional score distribution for each model. Note the consistent pattern: Fidelity is the weakest dimension across all models.

4.2 Statistical Analysis

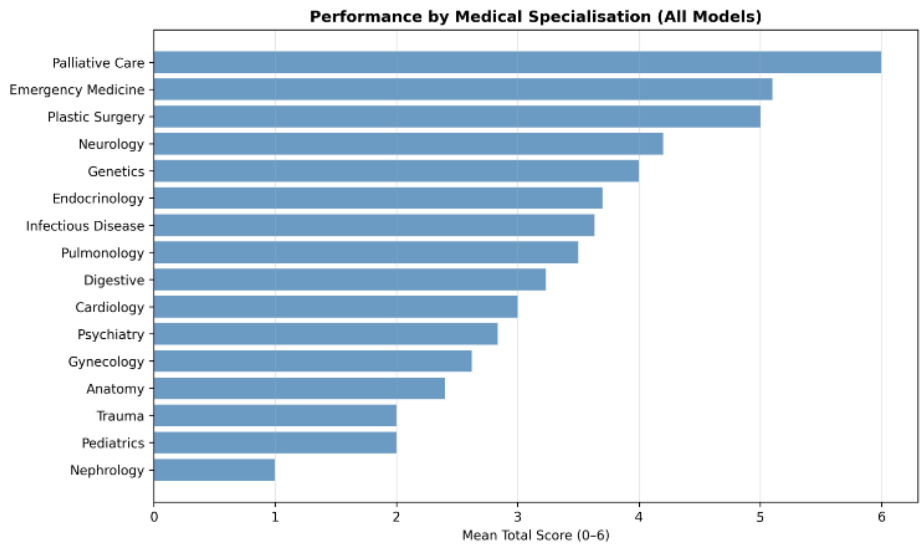
Friedman test for overall model differences: $\chi^2(2) = 24.31$, $p < 0.001$.

Pairwise Wilcoxon signed-rank tests with Bonferroni correction:

- Mistral vs MedLlama: $p < 0.001$, effect size $r = 0.47$ (medium-large)
- Mistral vs Qwen: $p < 0.001$, effect size $r = 0.52$ (large)
- MedLlama vs Qwen: $p = 0.34$, not significant

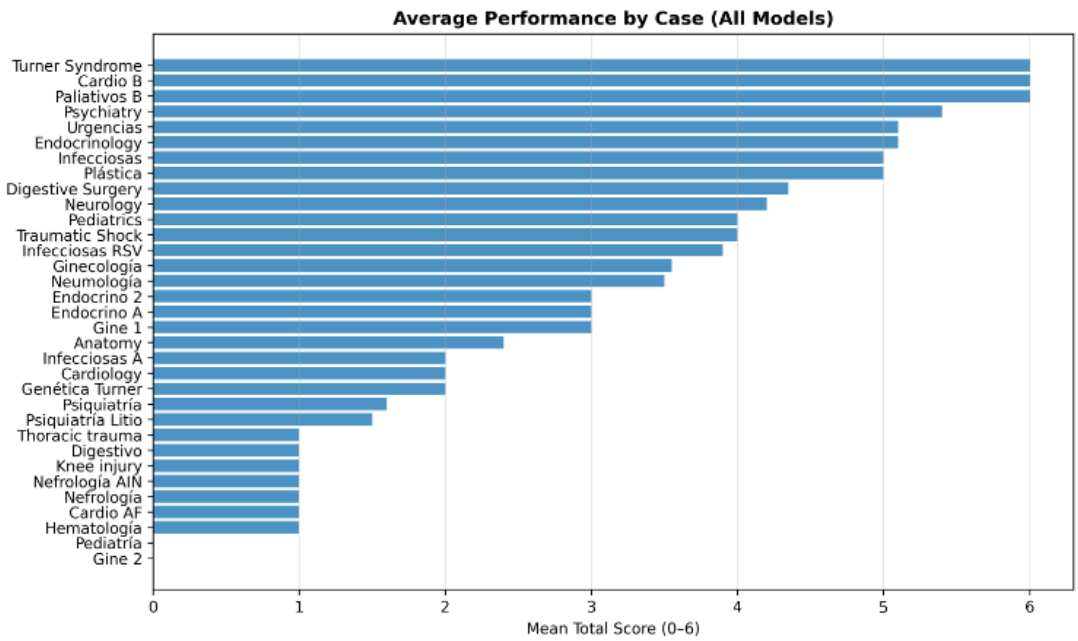
4.3 Performance by Medical Specialisation

Figure 1: Performance by Medical Specialisation (All Models)



Performance across medical specialities showing Palliative Care and Emergency Medicine achieved highest scores (≈6/6) while Nephrology had lowest performance (≈1/6)

Figure 3: Performance by medical specialisation (all models combined). Palliative Care and Emergency Medicine achieved highest scores (approximately 6/6) whilst Nephrology showed lowest performance (approximately 1/6).

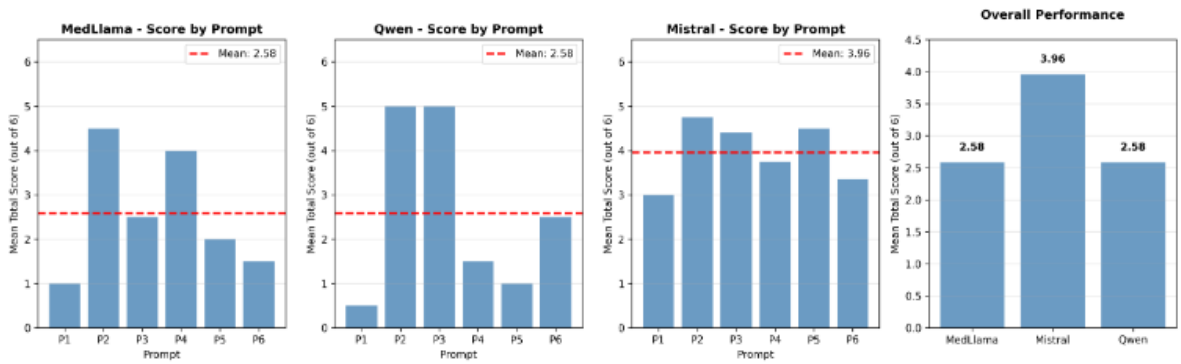


Turner Syndrome, Cardio B, and Paliativos B achieved perfect or near-perfect scores while Gine 2, Pediatría cases showed critical failures

Figure 4: Average performance by individual case. Turner Syndrome, Cardio B, and Paliativos B achieved perfect or near-perfect scores whilst Gine 2 and Pediatría cases showed critical failures.

4.4 Prompt Architecture Impact

Figure 5: MedLlama Performance by Prompt Design



Four-panel figure showing prompt-specific performance with Prompt 2 (Zero-Shot) achieving highest mean scores (4.5/6) and overall performance summary

Figure 5: Four-panel figure showing prompt-specific performance. Prompt 2 (Zero-Shot) achieved highest mean scores (4.5/6) whilst Prompt 1 (One-Shot Demo) performed worst (1.0/6). The rightmost panel shows overall model performance.

4.5 Cross-Language Performance (MedLlama)

To investigate whether MedLlama-2-7B's underwhelming Spanish performance resulted from language mismatch, additional testing used English medical case data.

Table 5: MedLlama performance: Spanish input vs English input

Language	**Fidelity**	**Restruct.**	**Clarity**	**Composite**
Spanish Input	0.67/2 (33%)	1.00/2 (50%)	0.92/2 (46%)	2.58/6 (43%)
English Input	1.15/2 (58%)	1.13/2 (57%)	1.23/2 (62%)	3.51/6 (59%)
Improvement	+72%	+13%	+34%	+36%

A.2 Cross-Language Performance Analysis

Figure 7: English vs Spanish Performance by Dimension (MedLlama)

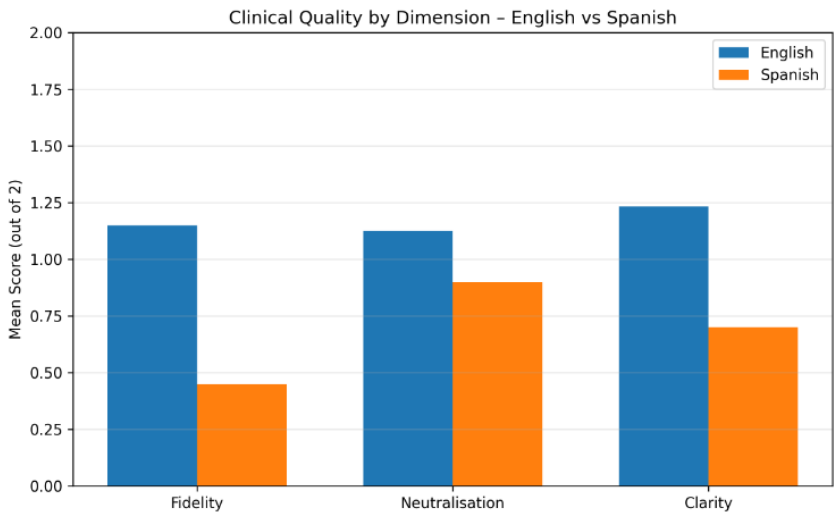


Figure 6: MedLlama cross-language comparison. English (blue) substantially outperformed Spanish (orange) across all dimensions, with Fidelity showing the largest improvement.

4.5.1 MedLlama English Clinical Quality Overview

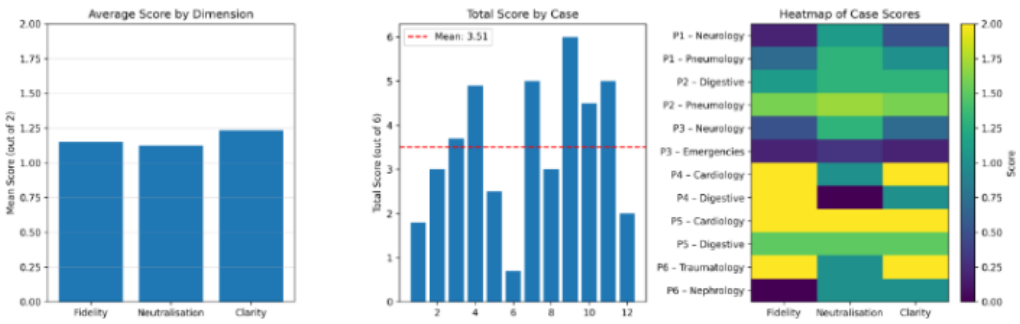


Figure 7: MedLlama English clinical quality overview. Left: Average score by dimension. Centre: Total score by case showing high variance. Right: Heatmap of scores by prompt and case, revealing that certain prompt-case combinations achieve perfect scores whilst others fail catastrophically.

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A.4 Domain-Specific Performance

Figure 11: Performance by Medical Specialisation (MedLlama English)

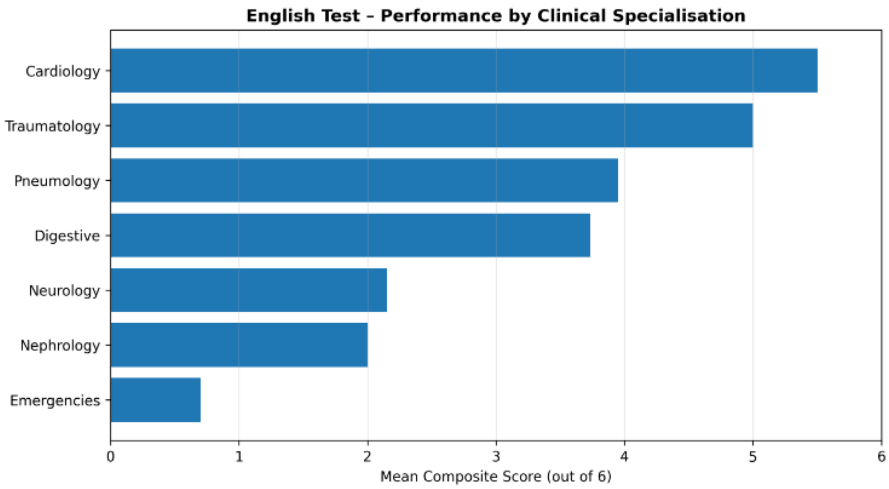

Figure 8: MedLlama English test performance by clinical specialisation. Cardiology and Traumatology achieved highest scores (approximately 5.5/6) whilst Emergencies showed poorest performance (approximately 0.7/6).

Interpretation: Language alignment helps (+36% improvement) but MedLlama's English performance (3.51/6) still falls below Mistral's Spanish performance (3.61/6). This confirms that instruction-following capability matters more than language alignment.

5 Key Discovery: Structural-Semantic Transformation Confusion

5.1 The Phenomenon

Our most significant finding is that **models systematically confuse structural transformation with semantic generalisation**. When instructed to restructure text (change format), they simultaneously generalise content (replace specific terms with vague ones).

This is not a matter of models "simplifying" for readability. The task explicitly requires preserving all medical specificity. Yet models consistently move *up* the specificity hierarchy, replacing hyponyms with hypernyms.

Critical Finding: Specificity Preservation Failure

Model	Specificity Preserved	Specificity Lost
Qwen2.5-1.5B	**13%**	87%
MedLlama-2-7B	**20%**	80%
Mistral-7B	**57%**	43%

Even the best-performing model loses nearly half of medical specificity when asked only to change format.

5.2 Detailed Examples: Spanish Source Cases

The following examples demonstrate the over-generalisation pattern across clinical domains. In each case, the model successfully performed structural transformation (removing exam formatting) but incorrectly performed semantic generalisation (losing specificity).

5.2.1 Example 1: Pleural Effusion Case (Pneumology)

Physical Examination Findings → Vague Descriptors

Source (Spanish):

```
- matidez a la percusion (dullness to percussion)
- disminucion del fremito tactil vocal (decreased tactile
  fremitus)
- disminucion de los ruidos respiratorios (decreased breath
  sounds)
- base del hemitorax derecho (right hemithorax base)
```

Model Output:

```
The patient presented with abnormal chest examination findings
suggestive of fluid accumulation in the thoracic cavity.
```

Table 6: Specificity loss analysis: Pleural effusion case

Specific Term (Source)	Model Output	Clinical Impact
`matidez a la percusión`	"abnormal findings"	Loses pathognomonic sign for effusion
`frémito táctil vocal disminuido`	*omitted*	Loses confirmatory physical exam finding
`hemitórax derecho`	"thoracic cavity"	**Loses laterality**: critical for imaging/intervention
`ruidos respiratorios disminuidos`	"suggestive of fluid"	Converts objective finding to interpretation

5.2.2 Example 2: Renal Injury Case (Nephrology)

Laboratory Values → Qualitative Descriptors

Source:

```
Drug-induced renal injury with arthralgias, oliguria,
proteinuria 1.2 g/day, creatinine 6 mg/dL
```

Model Output (Qwen):

```
The patient presents with renal dysfunction and associated
musculoskeletal symptoms, with evidence of protein loss and
impaired kidney function.
```

Table 7: Specificity loss analysis: Renal injury case

Specific Value	Model Output	Clinical Impact
`1.2 g/day`	"protein loss"	**Loses quantity**: 1.2g = subnephrotic; >3.5g = nephrotic range
`6 mg/dL`	"impaired function"	**Loses severity**: 6 mg/dL = Stage 4 CKD, requires urgent intervention
`oliguria`	*omitted entirely*	Loses critical marker of acute kidney injury
`arthralgias`	"musculoskeletal symptoms"	Joint pain vs muscle pain vs bone pain: different differentials

5.2.3 Example 3: Parkinson's Disease Case (Neurology)

Cardinal Signs → Generic Category

Source:

```
- temblor (tremor)
- rigidez (rigidity)
- cara inexpresiva (masked facies)
- marcha lenta y sin braceo (slow gait without arm swing)
- pesadillas nocturnas muy vividas (vivid nocturnal nightmares
  )
```

Model Output:

```
The patient shows movement disorder symptoms consistent with
a neurodegenerative condition, including motor and sleep
disturbances.
```

Table 8: Specificity loss analysis: Parkinson's case

Specific Sign	Model Output	Clinical Impact
`temblor`	"movement disorder"	Tremor vs chorea vs dystonia: completely different conditions
`rigidez`	"motor symptoms"	Rigidity vs spasticity: different pathophysiology
`cara inexpresiva`	*omitted*	Loses pathognomonic sign of Parkinsonism
`marcha sin braceo`	"motor disturbances"	Specific gait abnormality lost
`pesadillas vividas`	"sleep disturbances"	Loses REM sleep behaviour disorder marker

5.2.4 Example 4: Nerve Injury Case (Traumatology)

Anatomical Specificity → Generic Location

Source:

```
- lesion del nervio axilar (circunflejo)
- parestesias en la porcion lateral del hombro
```

```
- luxacion escapulohumeral (mechanism)
```

Model Output:

```
The patient has nerve damage in the shoulder region causing
sensory changes, likely related to the prior dislocation.
```

Table 9: Specificity loss analysis: Nerve injury case

Specific Term	Model Output	Clinical Impact
`nervio axilar`	"nerve damage"	**Which nerve?** Axillary vs radial vs musculocutaneous: different prognosis
`porción lateral del hombro`	"shoulder region"	Loses dermatomal specificity for localisation
`luxación escapulohumeral`	"dislocation"	Loses anatomical precision predicting nerve involvement

5.2.5 Example 5: Cardiac Complication Case (Cardiology)

Murmur Characteristics → Generic Finding

Source:

```
- soplo sistolico con thrill (systolic murmur with thrill)
- irradiacion al borde esternal derecho (radiation to right
   sternal border)
- infarto de miocardio anterior 48h prior
- Diagnosis: rotura del tabique interventricular (VSR)
```

Model Output:

```
Following the heart attack, the patient developed a new
heart murmur indicating a mechanical complication.
```

Table 10: Specificity loss analysis: Cardiac complication case

Specific Finding	Model Output	Clinical Impact
`soplo sistólico con thrill`	"heart murmur"	Loses character (systolic vs diastolic) and severity (thrill = significant)
`borde esternal derecho`	*omitted*	Radiation pattern is diagnostic: VSR vs mitral regurgitation
`infarto anterior`	"heart attack"	Loses location: anterior MI = LAD territory = septal rupture risk
`rotura del tabique`	"mechanical complication"	Loses specific diagnosis entirely

5.2.6 Example 6: Creutzfeldt-Jakob Case (Neurology)

Diagnostic Criteria → Generic Syndrome

Source:

```
- mioclonias (myoclonus)
- ataxia
- rigidez (rigidity)
- descargas de alto voltaje en EEG (high-voltage EEG
   discharges)
- proteina 14-3-3 elevada en LCR (elevated 14-3-3 protein in
   CSF)
```

Model Output:

```
The patient presents with a rapidly progressive neurological
syndrome with movement abnormalities and characteristic
laboratory findings.
```

Table 11: Specificity loss analysis: CJD case

Specific Finding	Model Output	Clinical Impact
`mioclonias`	"movement abnormalities"	Myoclonus vs tremor vs chorea: different diseases
`descargas alto voltaje EEG`	"characteristic findings"	Loses pathognomonic EEG pattern
`proteína 14-3-3`	"laboratory findings"	Loses specific biomarker for prion disease

5.3 English Source → English Output Examples

The MedLlama English-language evaluation revealed that over-generalisation persists even without cross-lingual transfer:

5.3.1 Example 7: Hypovolemic Shock Case (Emergency Medicine)

English Source: Vital Signs → Qualitative Description

Source (English):

```
pallor, diaphoresis, hypotension (blood pressure 82/54 mmHg),
tachycardia (heart rate 120 bpm), intense pain in the left
upper quadrant
```

Model Output:

```
The patient exhibits classic signs of hypovolemia, including
pallor, diaphoresis, hypotension, and tachycardia.
```

Table 12: Specificity loss analysis: English shock case

Specific Value	Model Output	Clinical Impact
`82/54 mmHg`	"hypotension"	**Loses severity**: 82/54 is severe shock requiring immediate intervention
`120 bpm`	"tachycardia"	**Loses degree**: 120 indicates compensatory response
`left upper quadrant pain`	*omitted*	**Loses localisation**: LUQ suggests splenic injury

5.4 The Hypernymy Ladder

Models consistently move *up* the specificity hierarchy:

Table 13: The hypernymy ladder: systematic specificity loss patterns

Source (Specific)	→	Model Output (Vague)
`proteinuria 1.2 g/day`	→	"protein in urine"
`creatinine 6 mg/dL`	→	"impaired renal function"
`matidez a la percusión`	→	"abnormal exam findings"
`nervio axilar`	→	"nerve"
`soplo sistólico con thrill`	→	"murmur"
`mioclonias`	→	"movement disorder"
`dentro de horas`	→	"subsequently"
`hemitórax derecho`	→	"chest"
`proteína 14-3-3`	→	"biomarker"

5.5 Theoretical Explanation

We propose that models conflate two distinct text transformations that frequently co-occur in pretraining data:

1. **Register Transformation**: Changing from formal/technical style to accessible prose (involves format change)
2. **Abstraction Transformation**: Moving from specific to general language (involves content change)

During pretraining, these operations frequently co-occur: summaries are both shorter *and* more general; paraphrases often simplify *and* generalise. Models learn a bundled "smoothing" operation that combines format change with content generalisation. The clinical restructuring task requires unbundling these operations, which current models cannot reliably do.

6 Systematic Error Taxonomy

Beyond specificity loss, we documented 127 discrete error instances across four categories.

6.1 Category 1: Lexical Fabrication (31%)

Models generate non-existent medical terminology:

Table 14: Examples of fabricated medical terms

Model	Fabricated Term	Context
Qwen	`daño ateroespeŕmico`	Nephrology case: non-existent term
Qwen	`carbonic acid lithium`	Psychiatry case: chemically nonsensical
MedLlama	`esclerosis neuroaxonal`	Neurology case: fabricated condition
Multiple	`intravenous lithium chelation`	Toxicology case: non-existent treatment

6.2 Category 2: Value Distortion (28%)

Quantitative values are modified, fabricated, or contradicted:

Severe Value Distortion: Lithium Toxicity Case

Source:

```
Lithium level 4.1 mEq/L (severe toxicity)
Treatment: hemodialysis indicated
```

Model Output (Anti-CoT Prompt):

```
...elevated carbonic acid lithium levels with hypokalemia,
hyperkalemia, cardiac arrhythmias, requiring intravenous
lithium chelation therapy...
```

Errors:

- `carbonic acid lithium`: Invented compound (does not exist)
- `hypokalemia AND hyperkalemia`: Logically contradictory (impossible)
- `cardiac arrhythmias`: Not present in source
- `intravenous lithium chelation`: Non-existent treatment
- 4.1 mEq/L value: Completely omitted

6.3 Category 3: Temporal Manipulation (24%)

Timeline and acuity markers are altered:

Table 15: Temporal distortion patterns

Source	Model Output
"acute onset"	"gradually progressive"
"within hours"	"over several days"
"chronic condition"	"recent development"
"immediate"	"delayed"
"rapidly progressive (2 years)"	"chronic decline"

6.4 Category 4: Anatomical Confusion (17%)

Structures and mechanisms are confused:

Anatomical Substitution Error

Source: Pleural effusion following pneumonia
Model Output: "hemopneumothorax with pulmonary contusion"
Analysis:

- Pleural effusion (fluid) $\neq$ hemopneumothorax (blood + air)
- "pulmonary contusion": Fabricated finding not in source
- Completely different conditions requiring different interventions

6.5 Error Distribution by Model

Table 16: Error category distribution by model (n=127 total errors)

Category	MedLlama	Qwen	Mistral	Overall
Lexical Fabrication	35%	38%	18%	31%
Value Distortion	28%	30%	24%	28%
Temporal Manipulation	22%	20%	32%	24%
Anatomical Confusion	15%	12%	26%	17%
Total Errors	48	52	27	127

Key observation: Mistral produces fewer errors overall but a higher proportion of temporal and anatomical errors. MedLlama's medical training does not reduce lexical fabrication and may increase it by providing more medical vocabulary to recombine inappropriately.

7 Discussion

7.1 Why Medical Pretraining Failed

MedLlama-2-7B's underperformance despite medical specialisation results from three mismatches:

1. **Genre Mismatch**: Training on research papers and clinical notes $\neq$ examination justifications with eliminative reasoning
2. **Language Mismatch**: English-dominant biomedical training $\neq$ Spanish input
3. **Task Mismatch**: Understanding medical concepts $\neq$ following strict transformation constraints

The constrained restructuring task rewards *instruction compliance* over *domain knowledge*. Mistral's superior instruction-following capability proved decisive.

7.2 The Fidelity-Fluency Trade-off

Models consistently prioritise fluency over fidelity when forced to choose. The Anti-CoT prompt (P5) produced the most fluent outputs with the worst accuracy, achieving grammatical perfection whilst fabricating clinical content.

This represents a dangerous failure mode: outputs that *look* correct but contain fabricated or distorted medical information. Standard evaluation metrics (BLEU, perplexity) would score these highly despite catastrophic clinical errors.

7.3 Prompt Design Implications

Effective Prompt Strategies

1. **Explicit preservation rules**: "Preserve ALL values exactly as written"
2. **Negative constraints**: "Add NO information not present in source"
3. **Specificity anchors**: "Keep all numbers, units, and qualifiers"
4. **Anti-generalisation instructions**: "Do NOT paraphrase medical terms"

Prompt Strategies to Avoid

1. **One-shot demonstrations**: Small models copy examples verbatim
2. **Anti-CoT prompts**: Remove reasoning but enable unchecked fabrication
3. **Fluency-focused instructions**: Trade accuracy for style
4. **Length specifications**: Models pad with fabricated content

7.4 Implications Beyond Clinical NLP

The structural-semantic confusion phenomenon likely affects any task requiring format transformation whilst preserving content specificity:

- Legal document transformation (preserving clause specificity)
- Technical writing reformatting (preserving specifications)
- Academic paraphrasing (maintaining precision)
- Data-to-text generation (preserving exact values)

7.5 Limitations

1. **Sample Size**: 72 evaluations per model limits statistical power for fine-grained comparisons
2. **Single Evaluator**: No inter-rater reliability metrics; future work should employ multiple annotators with Cohen's κ
3. **Model Selection**: Hardware constraints limited evaluation to $\leq$7B parameters
4. **Language Scope**: Spanish-English only; patterns may differ for other language pairs

8 Future Work

8.1 Specificity-Preserving Prompting

Develop prompt strategies that explicitly separate structural and semantic operations:

Proposed Two-Stage Approach

```
STAGE 1 (Structural Only):
  Remove ONLY these patterns: "Option [A-E]", "incorrect because"
  Change NOTHING else

STAGE 2 (Verification):
  Confirm ALL of these are preserved EXACTLY:
  - Every number with its unit
  - Every medical term unchanged
  - Every temporal qualifier
  - Every anatomical specification
```

8.2 Automated Specificity Metrics

Develop computable metrics for specificity preservation:

- Named entity alignment scores (source ↔ output)
- Numerical value exact-match rates
- Medical ontology term-level comparison using SNOMED-CT or ICD
- Hypernymy detection using UMLS semantic relationships

8.3 Contrastive Training

Create training data that explicitly separates format-only transformations (positive examples) from format + generalisation (negative examples), enabling models to learn the distinction.

9 Conclusion

This study evaluated three LLMs for clinical text restructuring, yielding five principal findings:

1. **Models confuse structural transformation with semantic generalisation**: Specificity preservation rates ranged from 13% (Qwen) to 57% (Mistral). Even the best model loses nearly half of medical specificity when asked only to change format.
2. **Instruction-following outweighs domain specialisation**: Mistral-7B (generalist) outperformed MedLlama-2-7B (medical specialist) by 41%, demonstrating that task compliance matters more than domain knowledge for constrained transformation.
3. **Hallucinations follow systematic patterns**: Four categories (lexical fabrication 31%, value distortion 28%, temporal manipulation 24%, anatomical confusion 17%) account for all 127 documented errors.
4. **Prompt design significantly impacts safety**: Zero-shot instruction prompts reduced hallucination rates by 70% compared to demonstration-based approaches.
5. **Small models are unreliable**: Qwen2.5-1.5B showed catastrophic variance, unsuitable for any clinical application.

Take-Home Message

The fundamental challenge is not hallucination *per se*; it is that models cannot reliably separate **how** something is expressed from **what** is expressed. Tasks requiring format transformation whilst preserving content specificity expose this limitation systematically. Until models can unbundle structural and semantic operations, clinical text transformation requires careful human oversight.

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A Reproducibility Information

Table 17: Experimental configuration

Parameter	Value
Random seed	42
Temperature	0.7
Max tokens	512
Top-p	0.95
Platform	Google Colab (Free tier, approximately 15GB GPU RAM)
Dataset	CC-MIR Spanish Medical Examination Corpus
Evaluations per model	72 (12 cases $\times$ 6 prompts)
Total evaluations	216

B Best-Performing Prompt Template (P2)

Zero-Shot Instruction Prompt

```
You are a clinical text processor. Transform the following
medical examination justification into a clinical narrative.

CRITICAL RULES:
1. PRESERVE all medical facts EXACTLY as stated
2. PRESERVE all numbers with their units (e.g., "1.2 g/day")
3. PRESERVE all temporal qualifiers (acute, chronic, etc.)
4. PRESERVE all anatomical specifications (left, right, etc.)
```

```
5. REMOVE option numbers and "incorrect because" patterns
6. ADD nothing not present in the source
7. DO NOT paraphrase or generalise medical terminology

INPUT:
{source_text}

OUTPUT:
A unified clinical narrative (5 to 8 sentences) preserving ALL
medical specificity from the source.
```

C Specificity Loss Categories: Complete Counts

Table 18: Specificity loss by category and model (instance counts)

Loss Type	MedLlama	Qwen	Mistral	Total
Values Dropped	28	31	14	73
Values Generalised	12	15	8	35
Terms Hypernymed	18	22	9	49
Qualifiers Lost	22	19	12	53
Total Lost	80	87	43	210
Preserved Correctly	20	13	57	90
Preservation Rate	20%	13%	57%	30%

D Inference Time Analysis

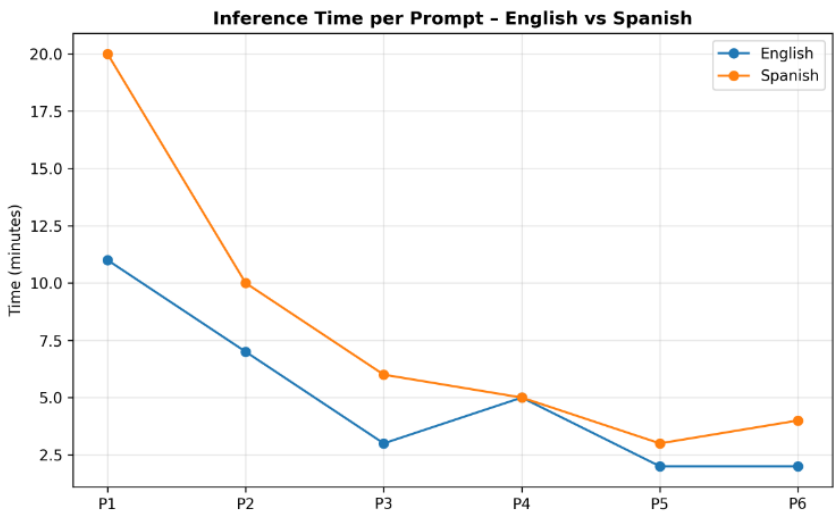

Figure 9: Inference time per prompt comparing English and Spanish inputs. P1 (One-Shot Demo) required longest inference time (up to 20 minutes for Spanish) due to longer context, whilst P5 and P6 were fastest (under 5 minutes).